Code No: 117FN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year I Semester Examinations, December-2023/January-2024
MOBILE COMPUTING
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Define mobile computing.	[2]
b)	Write a short note handover in GSM.	[3]
c)	What is the purpose of route optimization in mobile networks?	[2]
d)	What role does registration play in Mobile IP networks?	[3]
e)	What is database hoarding?	[2]
f)	List the advantages of Mobile TCP.	[3]
g)	What is a symmetric communication?	[2]
h)	Compare synchronous and asynchronous data transmission.	[3]
i)	Expand the acronym AODV and state its functionality.	[2]
j)	Explain the role of XML in mobile computing.	[3]

PART – B

(50 Marks)

2.a) Describe the significance of radio interfaces in GSM communication.
b) Discuss the paradigm and promises of mobile computing. [5+5]

OR

3.a) Differentiate between mobile communication and mobile computing.
b) Discuss about GSM services and how they contribute to efficient mobile communication. [5+5]

4.a) Elaborate on the concept of tunneling in Mobile IP networks.
b) Analyze the role of DHCP in mobile networks. [5+5]

OR

5.a) Explore how the design of a wireless MAC protocol can impact the quality of service in wireless networks.
b) Discuss the challenges and solutions related to packet delivery in wireless networks. [5+5]

6. Identify and elaborate on Quality of Service (QoS) issues specific to mobile database applications. How can these issues be addressed to ensure a better user experience? [10]

OR

7.a) How does adaptation play a role in mobile client-server communication? Explain.
b) Discuss the importance of backup and recovery strategies. [5+5]

8.a) Discuss the challenges associated with communications asymmetry.
b) Discuss the importance of protocols in ensuring smooth and reliable data synchronization. [5+5]

OR

9. Explain the various categories of data delivery mechanisms, providing examples for each. [10]

10. Explain the following:
a) Service discovery.
b) Linux for mobile devices. [5+5]

OR

11.a) State and explain the applications and challenges of a MANET.
b) Discuss the recent trends and advancements in Android as a mobile operating system. [5+5]

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